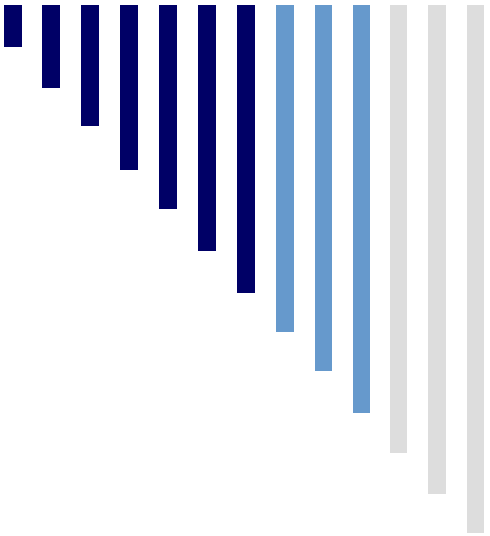


Statistics for Managers Using Microsoft® Excel

Regression

Learning Objectives

In this chapter, you learn:

- To use regression analysis to predict the value of a dependent variable based on an independent variable
- The meaning of the regression coefficients b_0 and b_1
- To evaluate the assumptions of regression analysis and know what to do if the assumptions are violated
- To make inferences about the slope and correlation coefficient
- To estimate mean values and predict individual values

Correlation vs. Regression

- A scatter plot (or scatter diagram) can be used to show the relationship between two numerical variables
- Correlation analysis is used to measure strength of the association (linear relationship) between two variables
  - Correlation is only concerned with strength of the relationship
  - No causal effect is implied with correlation

Regression Analysis

Regression analysis is used to:

- Predict the value of a dependent variable based on the value of at least one independent variable
- Explain the impact of changes in an independent variable on the dependent variable

Dependent variable: the variable you wish to explain

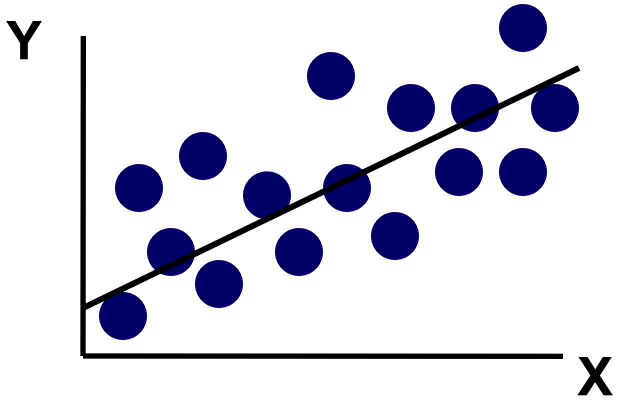
Independent variable: the variable used to explain the dependent variable

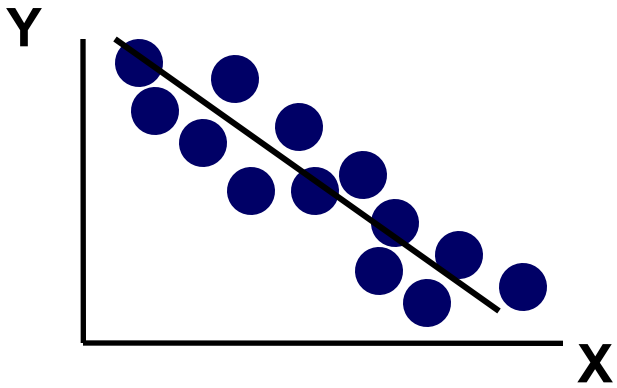
Simple Linear Regression Model

- Only **one** independent variable, X
- Relationship between X and Y is described by a linear function
- Changes in Y are related to changes in X

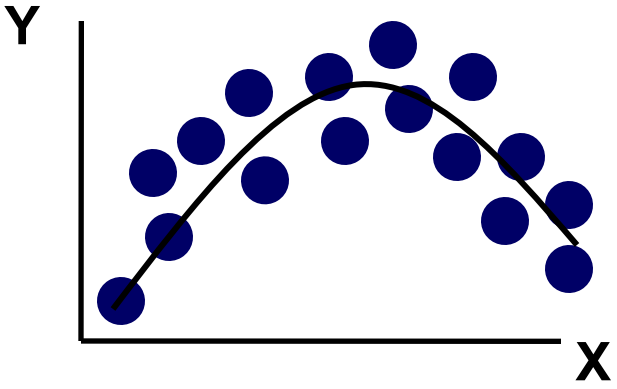
Types of Relationships

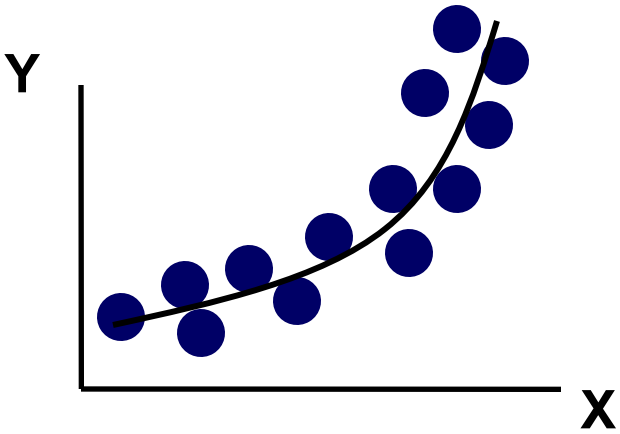
Linear relationships





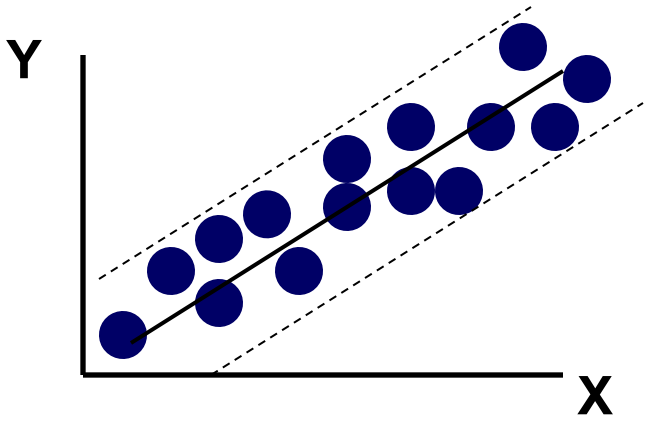
Curvilinear relationships

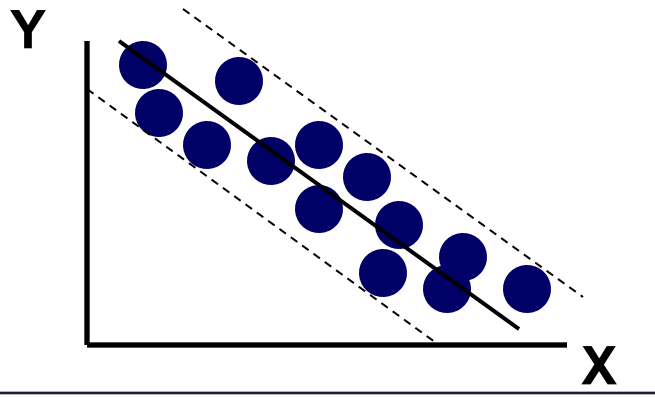




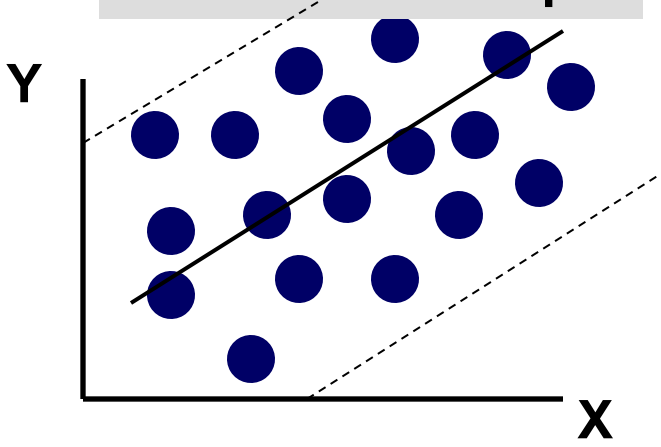
Types of Relationships

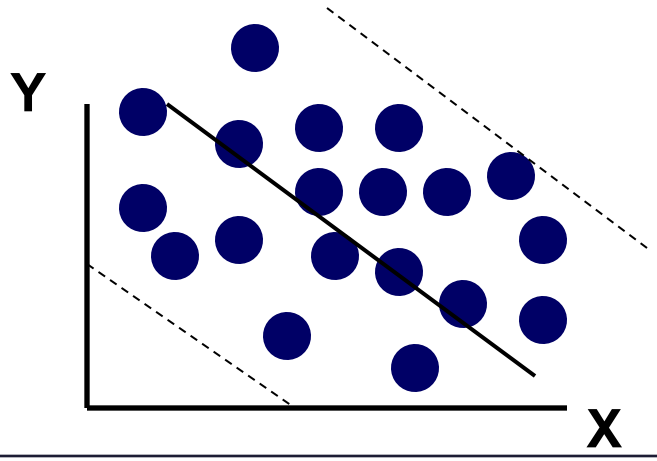
Strong relationships



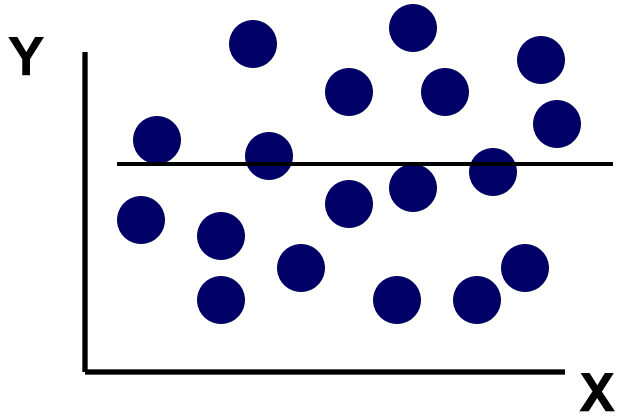


Weak relationships

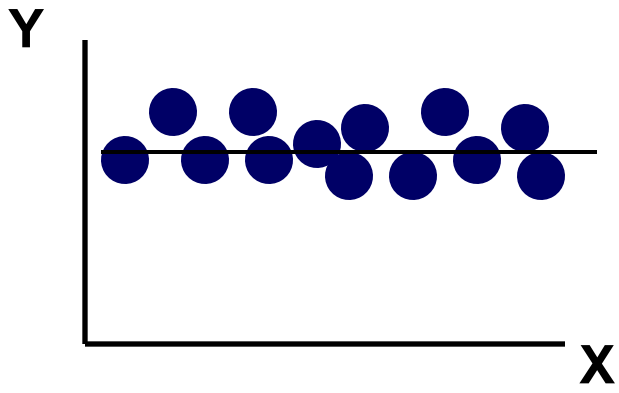




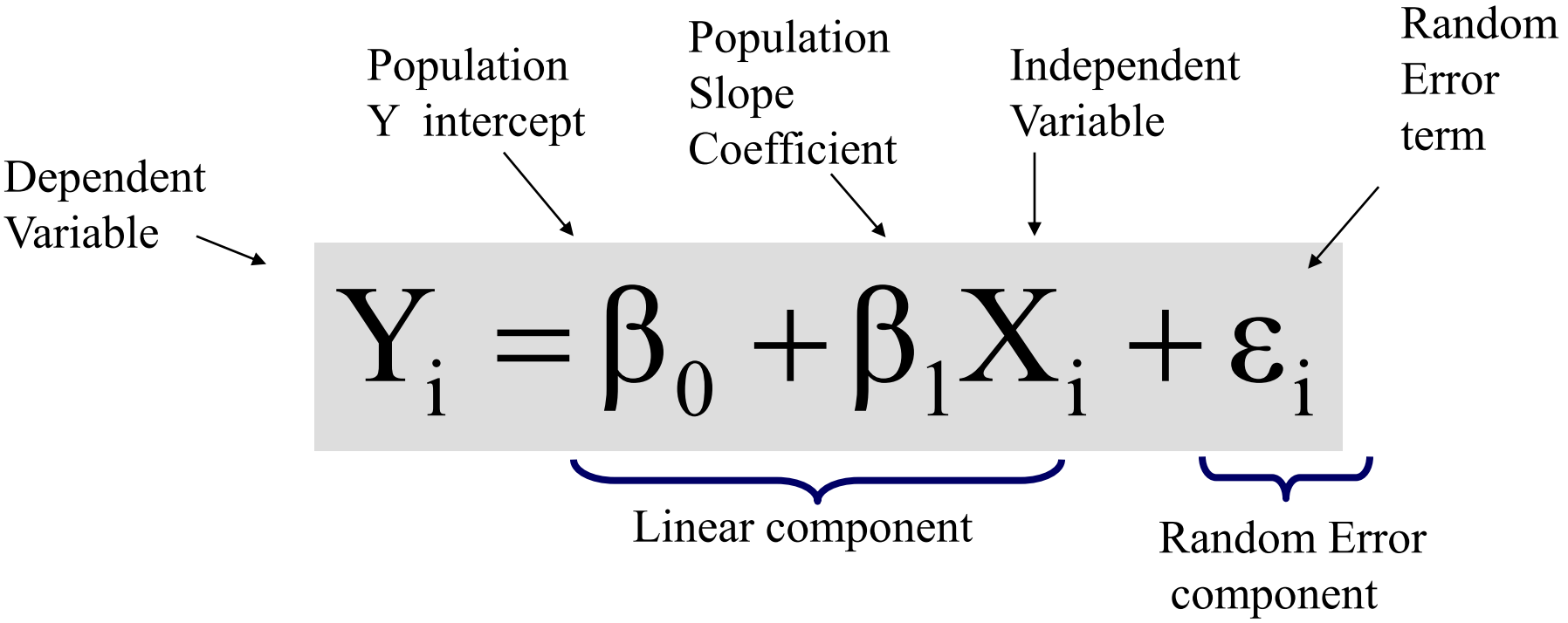
Types of Relationships



No relationship

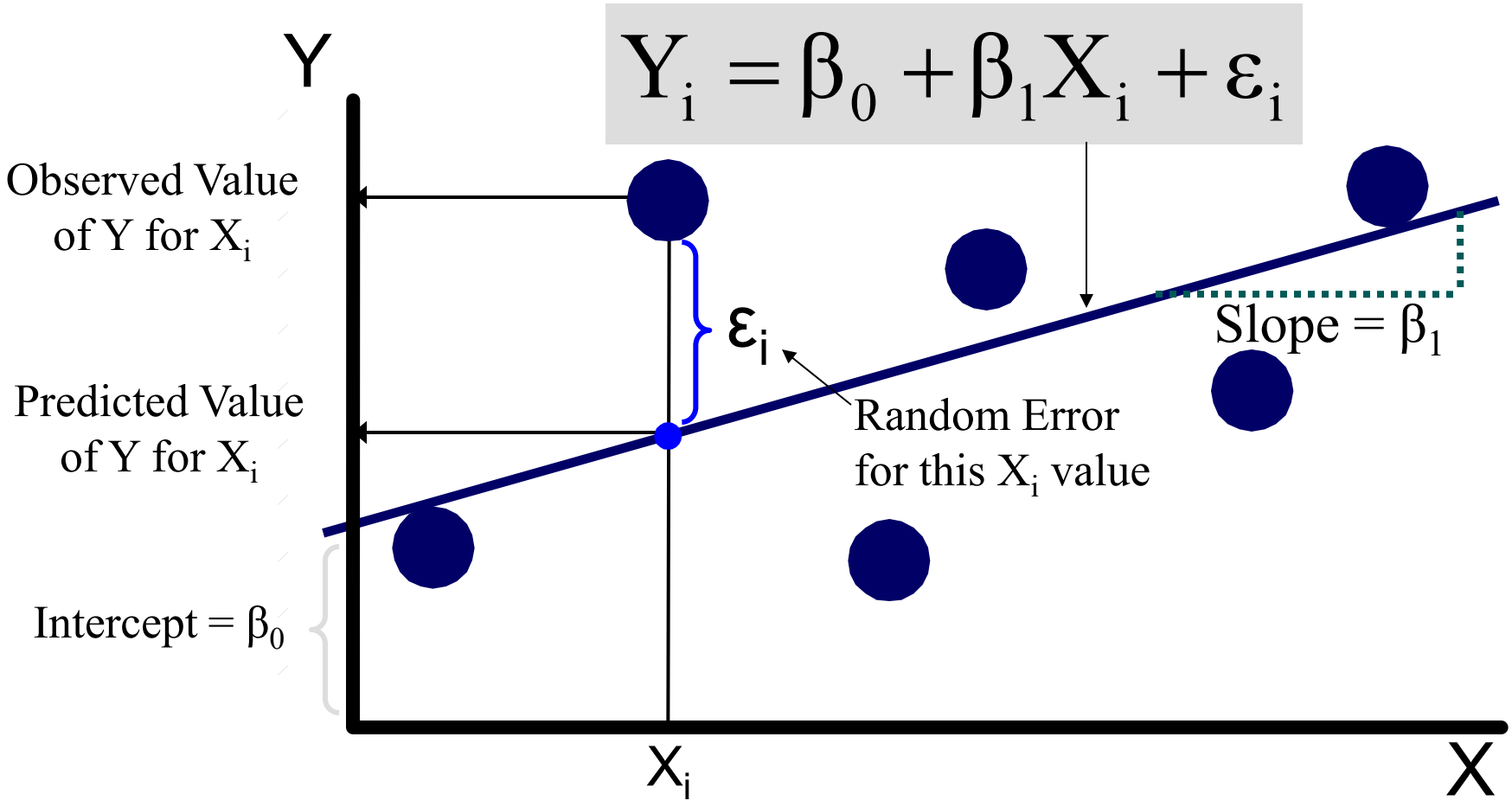


The Linear Regression Model



The population regression model:

The Linear Regression Model



Linear Regression Equation

The simple linear regression equation provides an estimate of the population regression line

Estimated (or predicted) Y value for observation i

Estimate of the regression intercept

Estimate of the regression slope

Value of X for observation i

$$\hat{Y}_i = b_0 + b_1 X_i$$

The Least Squares Method

- b_0 and b_1 are obtained by finding the values of b_0 and b_1 that minimize the sum of the squared differences between Y and $\hat{Y}$:

$$\min \sum (Y_i - \hat{Y}_i)^2 = \min \sum (Y_i - (b_0 + b_1 X_i))^2$$

Finding the Least Squares Equation

- The coefficients b_0 and b_1, and other regression results in this chapter, will be found using Excel

Formulas are shown in the text for those who are interested

Interpretation of the Intercept and the Slope

- b_0 is the estimated mean value of Y when the value of X is zero
- b_1 is the estimated change in the mean value of Y for every one-unit change in X

Linear Regression Example

- A real estate agent wishes to examine the relationship between the selling price of a home and its size (measured in square feet)

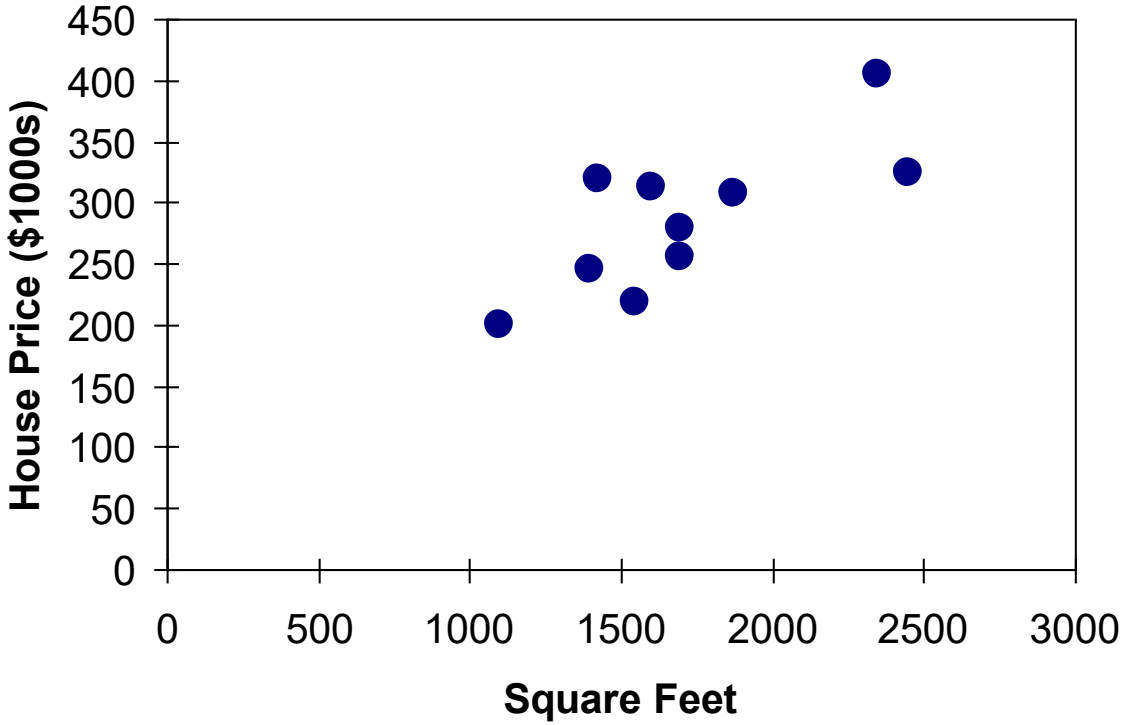
- A random sample of 10 houses is selected
  - Dependent variable (Y) = house price in $1000s
  - Independent variable (X) = square feet

Linear Regression Example Data

House Price in $1000s (Y)	Square Feet (X)
245	1400
312	1600
279	1700
308	1875
199	1100
219	1550
405	2350
324	2450
319	1425
255	1700

Linear Regression Example Scatterplot

- House price model: scatter plot

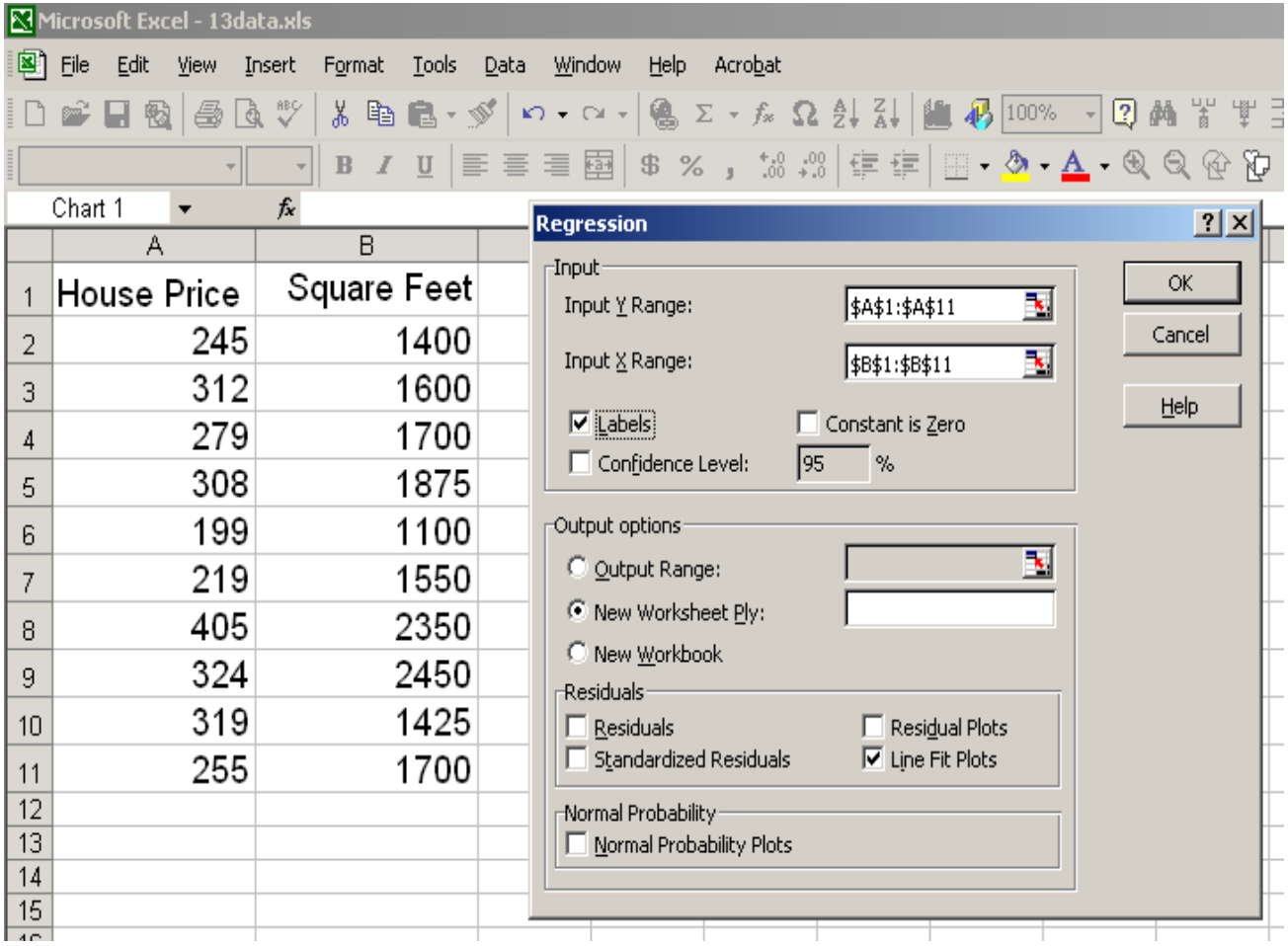


Linear Regression Example Using Excel

Tools

Data Analysis

Regression



	A	B
1	House Price	Square Feet
2	245	1400
3	312	1600
4	279	1700
5	308	1875
6	199	1100
7	219	1550
8	405	2350
9	324	2450
10	319	1425
11	255	1700
12		
13		
14		
15		

Linear Regression Example Excel Output

Regression Statistics	
Multiple R	0.76211
R Square	0.58082
Adjusted R Square	0.52842
Standard Error	41.33032
Observations	10

The regression equation is:

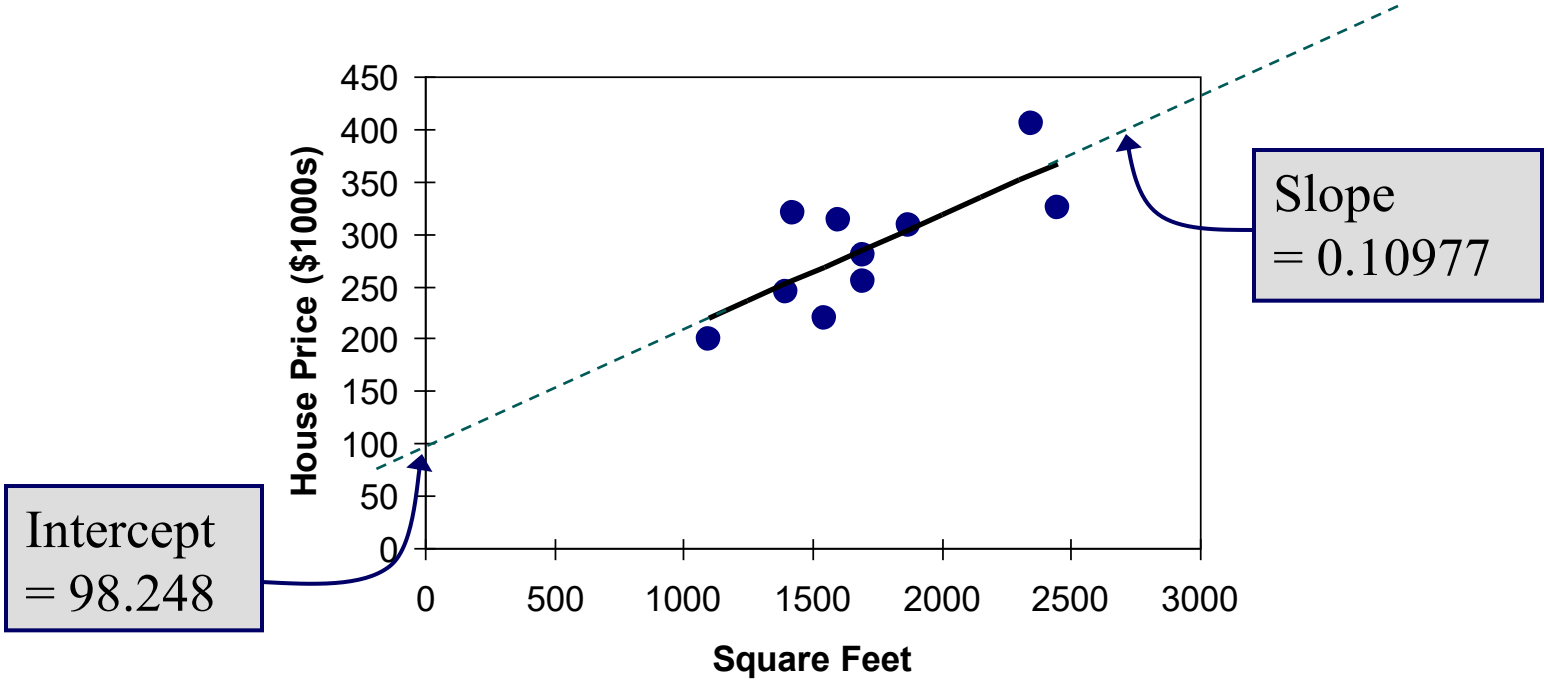
$$\text{house price} = 98.24833 + 0.10977\ (\text{square feet})$$

ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	1	18934.9348	18934.9348	11.0848	0.01039
Residual	8	13665.5652	1708.1957		
Total	9	32600.5000			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	98.24833	58.03348	1.69296	0.12892	-35.57720	232.07386
Square Feet	0.10977	0.03297	3.32938	0.01039	0.03374	0.18580

Linear Regression Example Graphical Representation

- House price model: scatter plot and regression line



$$\text{house price} = 98.24833 + 0.10977\ (\text{square feet})$$

Linear Regression Example Interpretation of b_0

$$\text{house price} = 98.24833 + 0.10977\ (\text{square feet})$$

- b_0 is the estimated mean value of Y when the value of X is zero (if X = 0 is in the range of observed X values)
- Because the square footage of the house cannot be 0, the Y intercept has no practical application.

Linear Regression Example Interpretation of b_1

$$\text{house price} = 98.24833 + 0.10977\,(\text{square feet})$$

- b_1 measures the mean change in the average value of Y as a result of a one-unit change in X
- Here, b_1 = .10977 tells us that the mean value of a house increases by .10977($1000) = $109.77, on average, for each additional one square foot of size

Linear Regression Example Making Predictions

Predict the price for a house with 2000 square feet:

$$\text{house price} = 98.25 + 0.1098\ (\text{sq.ft.})$$

$$= 98.25 + 0.1098(2000)$$

$$= 317.85$$

The predicted price for a house with 2000 square feet is 317.85($1,000s) = $317,850

Measures of Variation

Total variation is made up of two parts:

$$SST = SSR + SSE$$

Total Sum of Squares	Regression Sum of Squares	Error Sum of Squares
$SST = \sum (Y_i - \bar{Y})^2$	$SSR = \sum (\hat{Y}_i - \bar{Y})^2$	$SSE = \sum (Y_i - \hat{Y}_i)^2$

where:

$\bar{Y}$ = Mean value of the dependent variable

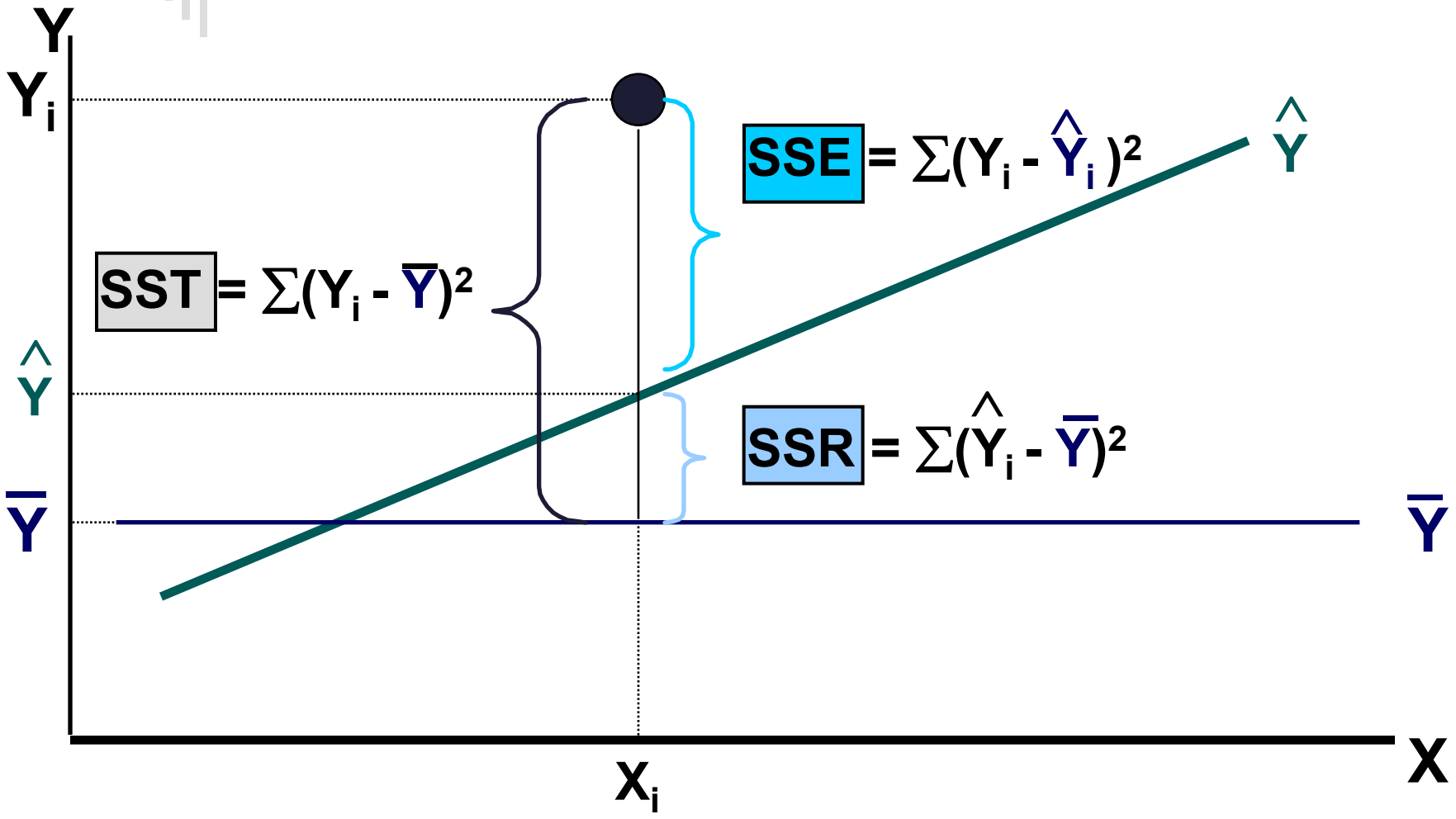
Y_i = Observed values of the dependent variable

$\hat{Y}_i$ = Predicted value of Y for the given X_i value

Measures of Variation

- SST = total sum of squares
  - Measures the variation of the Y_i values around their mean Y
- SSR = regression sum of squares
  - Explained variation attributable to the relationship between X and Y
- SSE = error sum of squares
  - Variation attributable to factors other than the relationship between X and Y

Measures of Variation



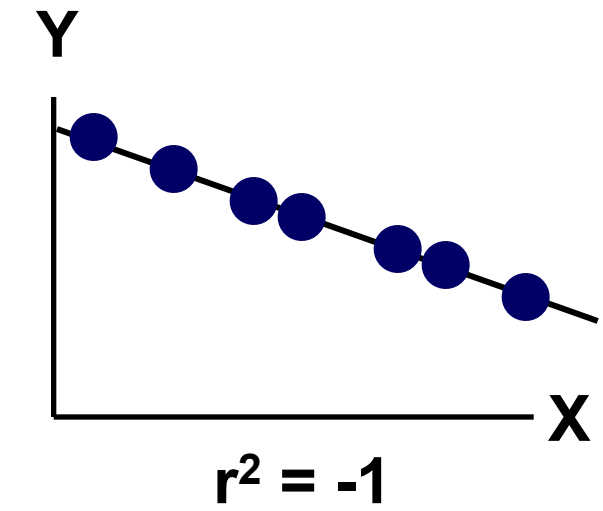
Coefficient of Determination, r^2

- The coefficient of determination is the portion of the total variation in the dependent variable that is explained by variation in the independent variable
- The coefficient of determination is also called r-squared and is denoted as r^2

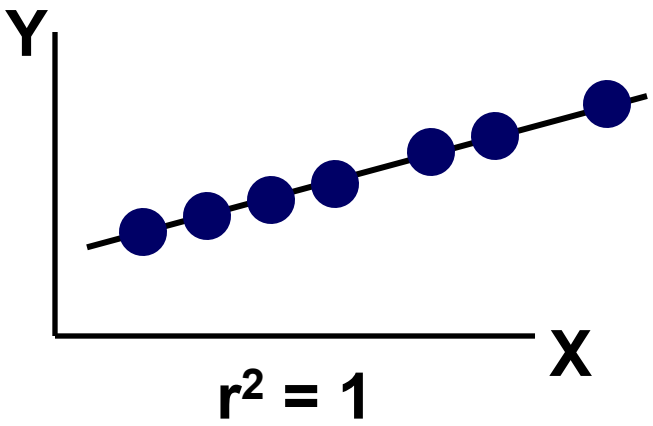
$$r^2 = \frac{SSR}{SST} = \frac{\text{regression sum of squares}}{\text{total sum of squares}}$$

$$0 \le r^2 \le 1$$

Coefficient of Determination, r^2



$r^2 = -1$



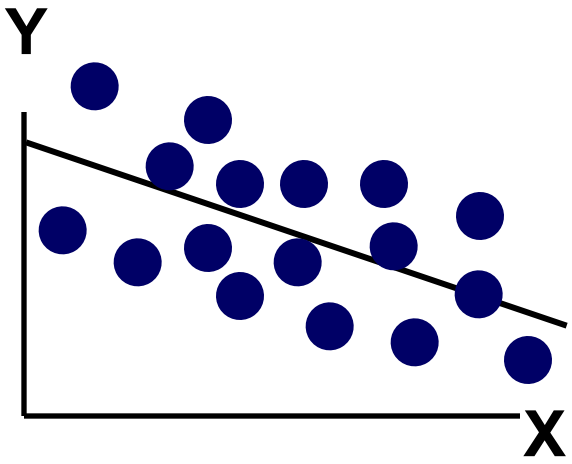
$r^2 = 1$

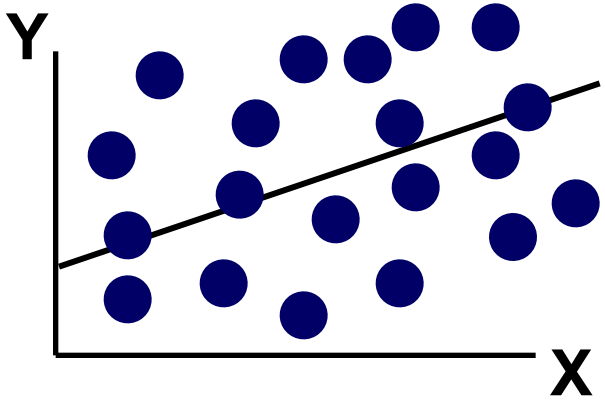
$r^2 = 1$

Perfect linear relationship between X and Y:

100% of the variation in Y is explained by variation in X

Coefficient of Determination, r^2



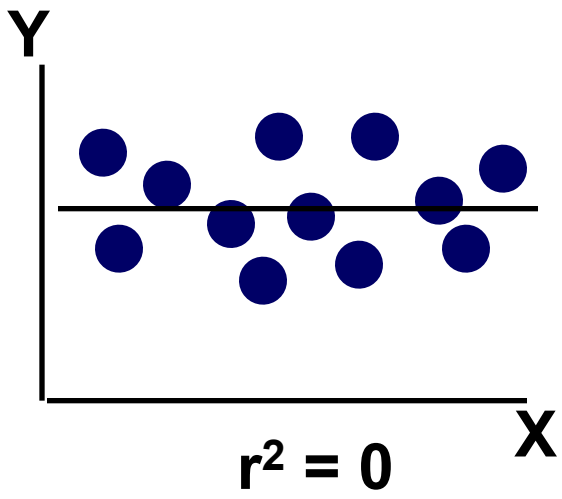


$0 < r^2 < 1$

Weaker linear relationships between X and Y:

Some but not all of the variation in Y is explained by variation in X

Coefficient of Determination, r^2



$r^2 = 0$

No linear relationship between X and Y:

The value of Y is not related to X. (None of the variation in Y is explained by variation in X)

Linear Regression Example Coefficient of Determination, r^2

Regression Statistics	
Multiple R	0.76211
R Square	0.58082
Adjusted R Square	0.52842
Standard Error	41.33032
Observations	10

$$r^2 = \frac{SSR}{SST} = \frac{18934.9348}{32600.5000} = 0.58082$$

58.08% of the variation in house prices is explained by variation in square feet

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	18934.9348	18934.9348	11.0848	0.01039
Residual	8	13665.5652	1708.1957		
Total	9	32600.5000			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	98.24833	58.03348	1.69296	0.12892	-35.57720	232.07386
Square Feet	0.10977	0.03297	3.32938	0.01039	0.03374	0.18580

Standard Error of Estimate

- The standard deviation of the variation of observations around the regression line is estimated by

$$S_{YX} = \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2}{n-2}}$$

Where

SSE = error sum of squares

n = sample size

Linear Regression Example Standard Error of Estimate

Regression Statistics	
Multiple R	0.76211
R Square	0.58082
Adjusted R Square	0.52842
Standard Error	41.33032
Observations	10

$S_{YX} = 41.33032$

ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	1	18934.9348	18934.9348	11.0848	0.01039
Residual	8	13665.5652	1708.1957		
Total	9	32600.5000			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	98.24833	58.03348	1.69296	0.12892	-35.57720	232.07386
Square Feet	0.10977	0.03297	3.32938	0.01039	0.03374	0.18580

Inferences About the Slope: t Test

- t test for a population slope
  - Is there a linear relationship between X and Y?
- Null and alternative hypotheses
  - H_0: $\beta_1 = 0$ (no linear relationship)
  - H_1: $\beta_1 \neq 0$ (linear relationship does exist)
- Test statistic

$$t = \frac{b_1 - \beta_1}{S_{b_1}}$$

$$d.f. = n - 2$$

where:

b_1 = regression slope coefficient

β_1 = hypothesized slope

S_{b1} = standard error of the slope

Inferences About the Slope: t Test Example

House Price in $1000s (y)	Square Feet (x)
245	1400
312	1600
279	1700
308	1875
199	1100
219	1550
405	2350
324	2450
319	1425
255	1700

Estimated Regression Equation:

$$\text{house price} = 98.25 + 0.1098\,(\text{sq.ft.})$$

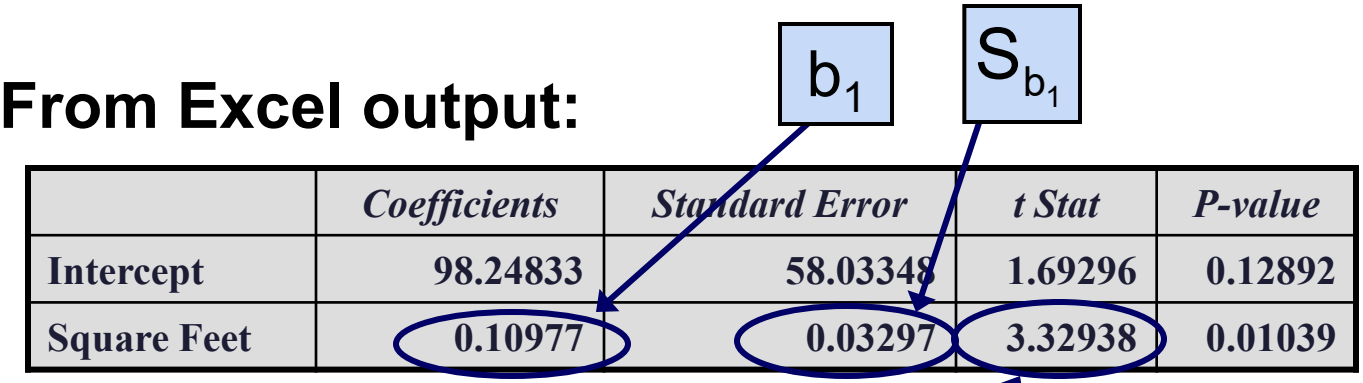
The slope of this model is 0.1098

Is there a relationship between the square footage of the house and its sales price?

Inferences About the Slope: t Test Example

- $H_0: \beta_1 = 0$
- $H_1: \beta_1 \neq 0$

From Excel output:



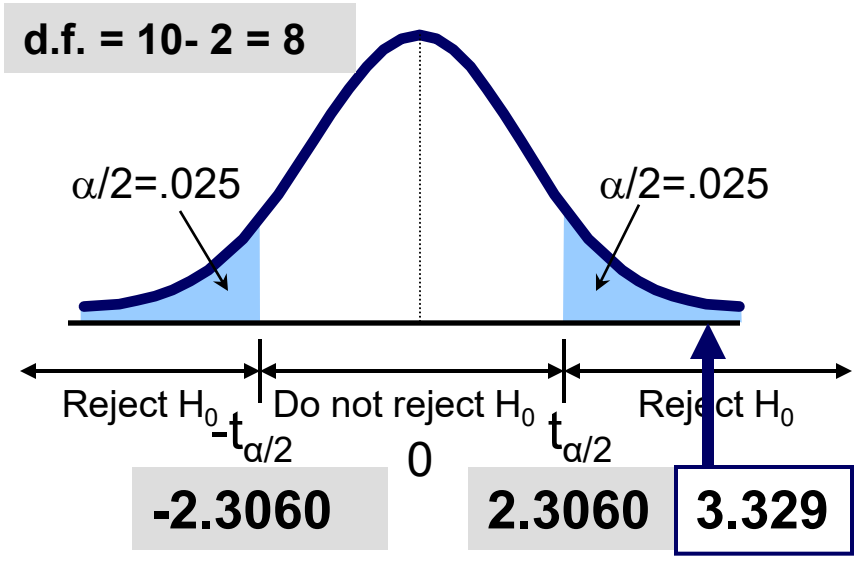
	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	98.24833	58.03348	1.69296	0.12892
Square Feet	0.10977	0.03297	3.32938	0.01039

$$t = \frac{b_1 - \beta_1}{S_{b_1}} = \frac{0.10977 - 0}{0.03297} = 3.32938$$

Inferences About the Slope: t Test Example

Test Statistic: **t = 3.329**

- $H_0: \beta_1 = 0$
- $H_1: \beta_1 \neq 0$



Decision: Reject H_0

There is sufficient evidence that square footage affects house price

Inferences About the Slope: t Test Example

From Excel output:

P-Value

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	**98.24833**	**58.03348**	**1.69296**	**0.12892**
Square Feet	**0.10977**	**0.03297**	**3.32938**	**0.01039**

- $H_0: \beta_1 = 0$
- $H_1: \beta_1 \neq 0$

Decision: Reject H_0, since p-value $< \alpha$

There is sufficient evidence that square footage affects house price.

F-Test for Significance

- F Test statistic:

$$F = \frac{MSR}{MSE}$$

where

$$MSR = \frac{SSR}{k}$$

$$MSE = \frac{SSE}{n - k - 1}$$

where F follows an F distribution with k numerator degrees of freedom and (n - k - 1) denominator degrees of freedom

(k = the number of independent variables in the regression model)

F-Test for Significance Excel Output

Regression Statistics	
Multiple R	0.76211
R Square	0.58082
Adjusted R Square	0.52842
Standard Error	41.33032
Observations	10

$$F = \frac{MSR}{MSE} = \frac{18934.9348}{1708.1957} = 11.0848$$

With 1 and 8 degrees of freedom

P-value for the F-Test

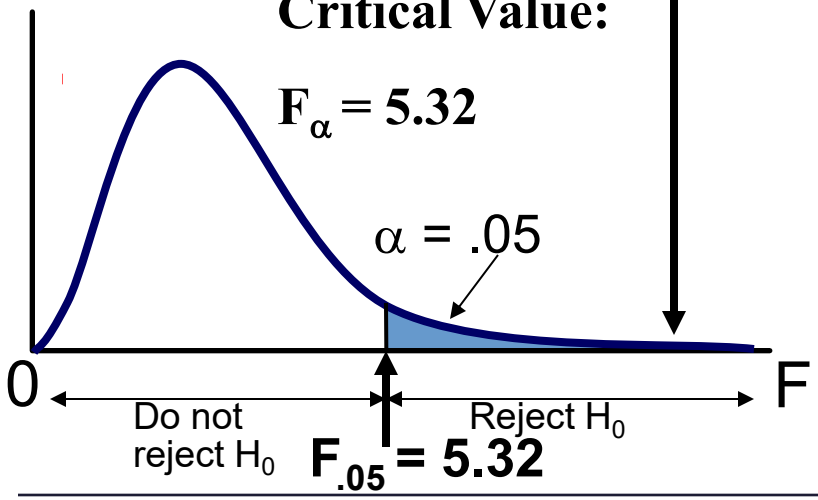
ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	1	18934.9348	18934.9348	11.0848	0.01039
Residual	8	13665.5652	1708.1957		
Total	9	32600.5000			

F-Test for Significance

- $H_0: \beta_1 = 0$
- $H_1: \beta_1 \neq 0$
- $\alpha = .05$
- $df_1 = 1 \qquad df_2 = 8$

Critical Value:

$F_\alpha = 5.32$



Test Statistic:

$$F = \frac{MSR}{MSE} = 11.08$$

Decision:

Reject H_0 at $\alpha = 0.05$

Conclusion:

There is sufficient evidence that house size affects selling price

Confidence Interval Estimate for the Slope

Confidence Interval Estimate of the Slope:

$$b_1 \pm t_{n-2} S_{b_1}$$

d.f. = n - 2

Excel Printout for House Prices:

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	**98.24833**	**58.03348**	**1.69296**	**0.12892**	**-35.57720**	**232.07386**
Square Feet	**0.10977**	**0.03297**	**3.32938**	**0.01039**	**0.03374**	**0.18580**

At the 95% level of confidence, the confidence interval for the slope is (0.0337, 0.1858)

Confidence Interval Estimate for the Slope

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	**98.24833**	**58.03348**	**1.69296**	**0.12892**	**-35.57720**	**232.07386**
Square Feet	**0.10977**	**0.03297**	**3.32938**	**0.01039**	**0.03374**	**0.18580**

Since the units of the house price variable is $1000s, you are 95% confident that the mean change in sales price is between $33.74 and $185.80 per square foot of house size

This 95% confidence interval does not include 0.

Conclusion: There is a significant relationship between house price and square feet at the .05 level of significance

t Test for a Correlation Coefficient

- Hypotheses
  - H_0: $\rho = 0$ (no correlation between X and Y)
  - H_1: $\rho \neq 0$ (correlation exists)

- Test statistic

$$t = \frac{r - \rho}{\sqrt{\frac{1 - r^2}{n - 2}}}$$

(with n – 2 degrees of freedom)

where

$r = +\sqrt{r^2}$ if $b_1 > 0$

$r = -\sqrt{r^2}$ if $b_1 < 0$

t Test for a Correlation Coefficient

Is there evidence of a linear relationship between square feet and house price at the .05 level of significance?

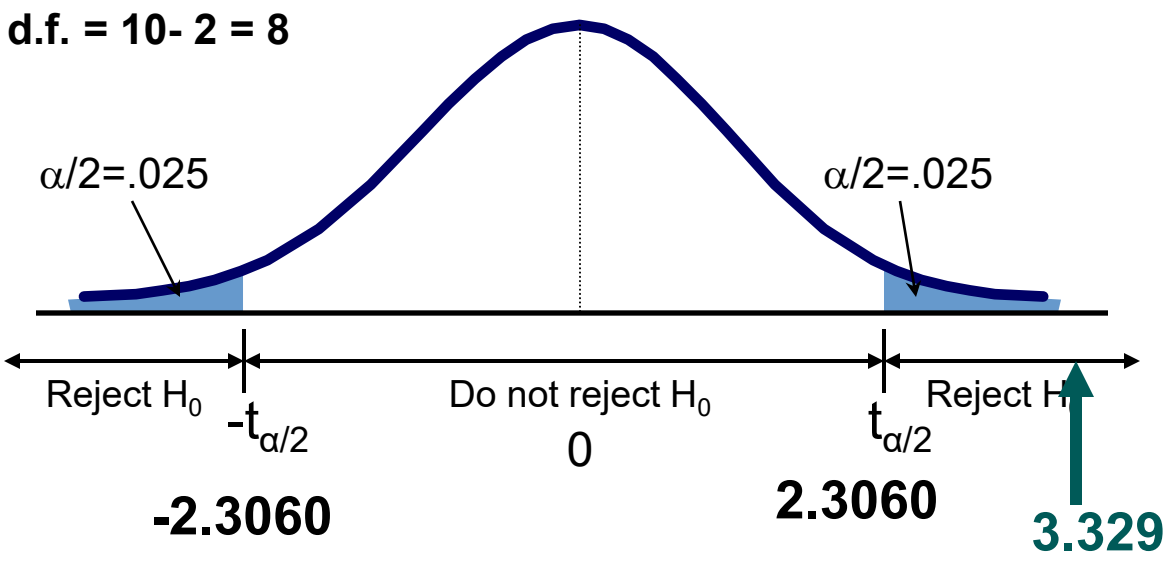
$H_0: \rho = 0$ (No correlation)

$H_1: \rho \neq 0$ (correlation exists)

$\alpha = .05$, $df = 10 - 2 = 8$

$$t = \frac{r - \rho}{\sqrt{\frac{1 - r^2}{n - 2}}} = \frac{.762 - 0}{\sqrt{\frac{1 - .762^2}{10 - 2}}} = 3.329$$

t Test for a Correlation Coefficient



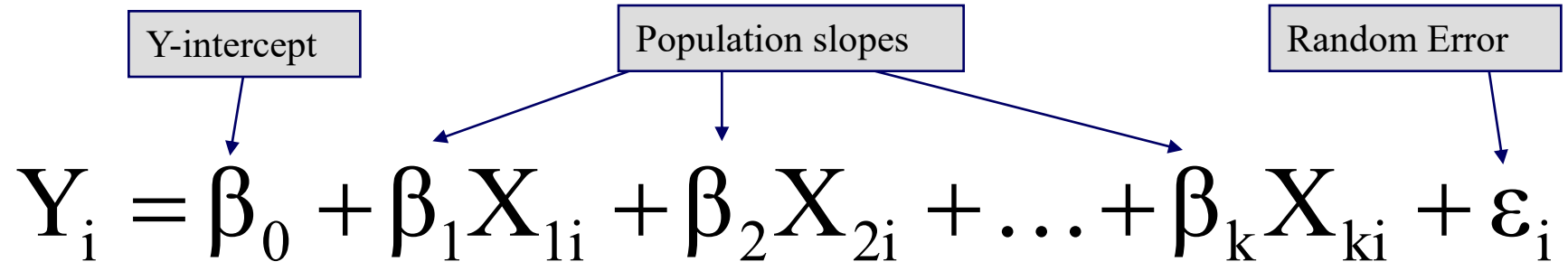
Decision:
Reject H_0

Conclusion:
There is evidence of a linear association at the 5% level of significance

The Multiple Regression Model

Idea: Examine the linear relationship between 1 dependent (Y) & 2 or more independent variables (X_i).

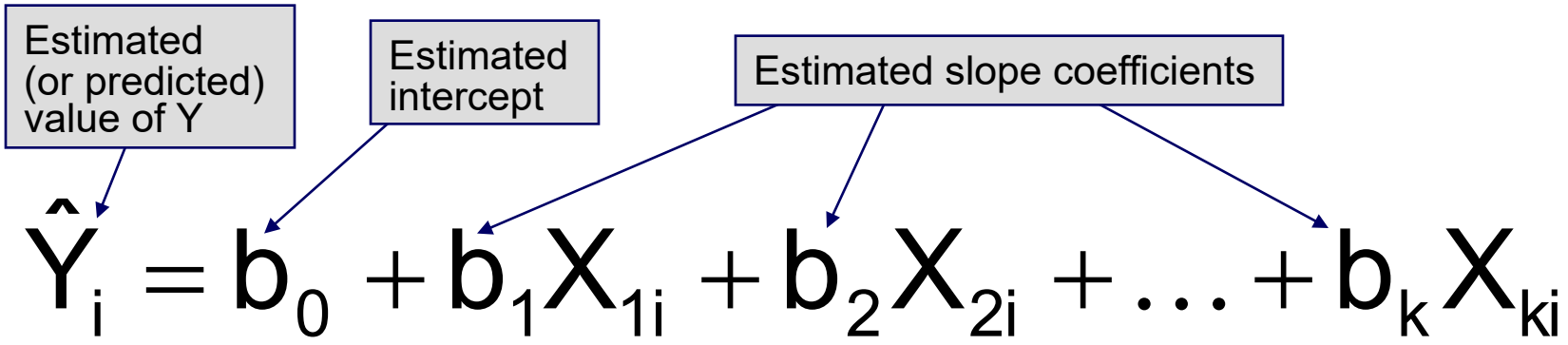
Multiple Regression Model with k Independent Variables:



Multiple Regression Equation

The coefficients of the multiple regression model are estimated using sample data

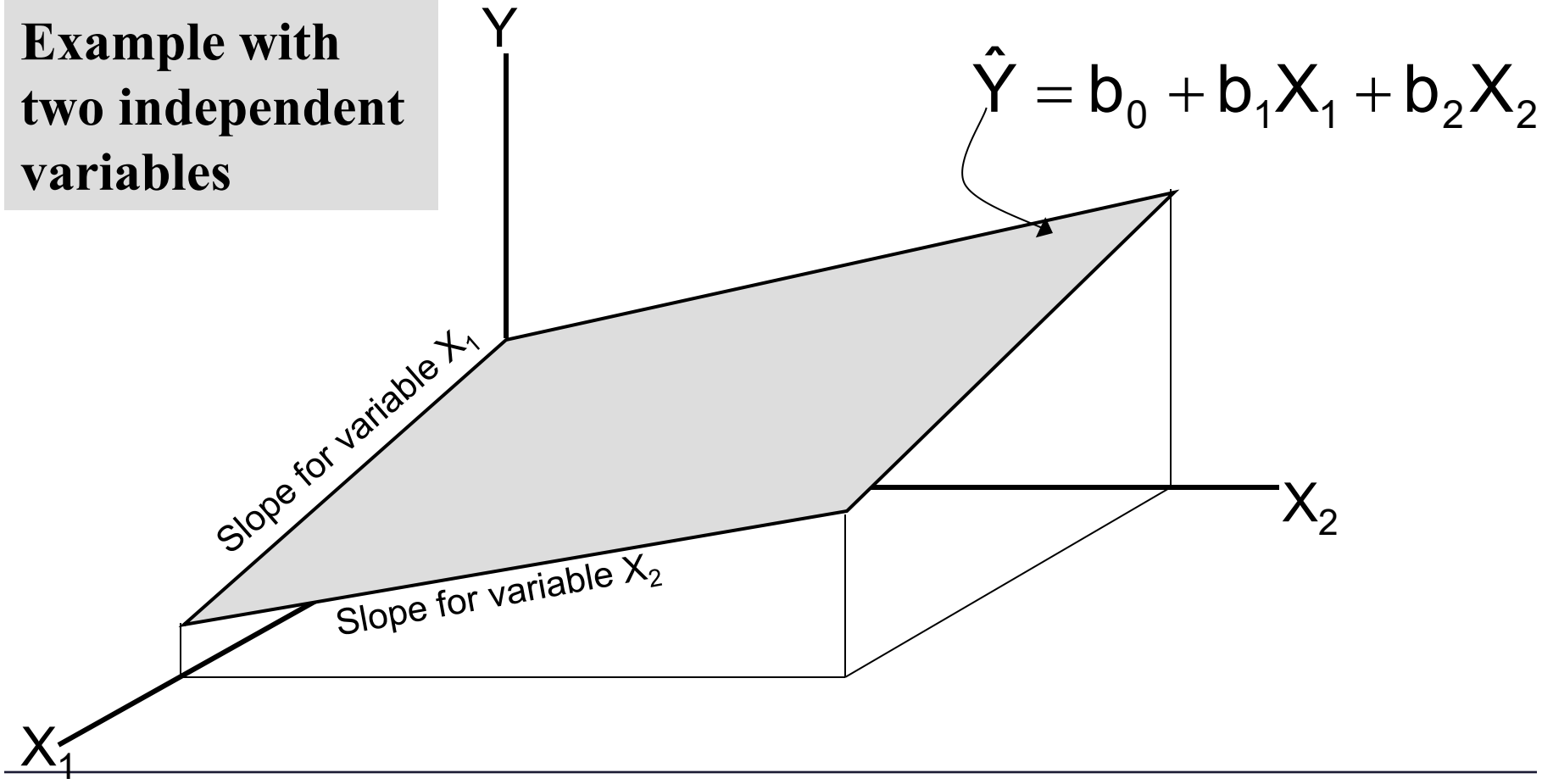
Multiple regression equation with k independent variables:



In this chapter we will always use Excel to obtain the regression slope coefficients and other regression summary measures.

Multiple Regression Equation

Example with two independent variables



Multiple Regression Equation 2 Variable Example

- A distributor of frozen dessert pies wants to evaluate factors thought to influence demand
  - Dependent variable: Pie sales (units per week)
  - Independent variables: Price (in $)
    Advertising ($100's)
- Data are collected for 15 weeks

Multiple Regression Equation 2 Variable Example

Week	Pie Sales	Price ($)	Advertising ($100s)
1	350	5.50	3.3
2	460	7.50	3.3
3	350	8.00	3.0
4	430	8.00	4.5
5	350	6.80	3.0
6	380	7.50	4.0
7	430	4.50	3.0
8	470	6.40	3.7
9	450	7.00	3.5
10	490	5.00	4.0
11	340	7.20	3.5
12	300	7.90	3.2
13	440	5.90	4.0
14	450	5.00	3.5
15	300	7.00	2.7

Multiple regression equation:

- $\text{Sales} = b_0 + b_1\,(\text{Price}) + b_2\,(\text{Advertising})$
- $\text{Sales} = b_0 + b_1X_1 + b_2X_2$

Where X_1 = Price

X_2 = Advertising

Multiple Regression Equation 2 Variable Example, Excel

Regression Statistics	
Multiple R	0.72213
R Square	0.52148
Adjusted R Square	0.44172
Standard Error	47.46341
Observations	15

$$\text{Sales} = 306.526 - 24.975(X_1) + 74.131(X_2)$$

ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	2	29460.027	14730.013	6.53861	0.01201
Residual	12	27033.306	2252.776		
Total	14	56493.333			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	306.52619	114.25389	2.68285	0.01993	57.58835	555.46404
Price	-24.97509	10.83213	-2.30565	0.03979	-48.57626	-1.37392
Advertising	74.13096	25.96732	2.85478	0.01449	17.55303	130.70888

Multiple Regression Equation 2 Variable Example

$$\text{Sales} = 306.526 - 24.975(X_1) + 74.131(X_2)$$

where

Sales is in number of pies per week
Price is in $
Advertising is in $100's.

b_1 = -24.975: sales will decrease, on average, by 24.975 pies per week for each $1 increase in selling price, net of the effects of changes due to advertising

b_2 = 74.131: sales will increase, on average, by 74.131 pies per week for each $100 increase in advertising, net of the effects of changes due to price

Multiple Regression Equation 2 Variable Example

Predict sales for a week in which the selling price is $5.50 and advertising is $350:

$$\begin{aligned} \text{Sales} &= 306.526 - 24.975(X_1) + 74.131(X_2) \\ &= 306.526 - 24.975(5.50) + 74.131(3.5) \\ &= 428.62 \end{aligned}$$

Predicted sales is 428.62 pies

Note that Advertising is in $100's, so $350 means that $X_2 = 3.5$

Coefficient of Multiple Determination

- Reports the proportion of total variation in Y explained by all X variables taken together

$$r^2 = \frac{\text{SSR}}{\text{SST}} = \frac{\text{regression sum of squares}}{\text{total sum of squares}}$$

Coefficient of Multiple Determination (Excel)

Regression Statistics	
Multiple R	0.72213
R Square	0.52148
Adjusted R Square	0.44172
Standard Error	47.46341
Observations	15

$$r^2 = \frac{SSR}{SST} = \frac{29460.0}{56493.3} = .52148$$

52.1% of the variation in pie sales is explained by the variation in price and advertising

ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	2	29460.027	14730.013	6.53861	0.01201
Residual	12	27033.306	2252.776		
Total	14	56493.333			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	306.52619	114.25389	2.68285	0.01993	57.58835	555.46404
Price	-24.97509	10.83213	-2.30565	0.03979	-48.57626	-1.37392
Advertising	74.13096	25.96732	2.85478	0.01449	17.55303	130.70888

Adjusted r^2

- r^2 never decreases when a new X variable is added to the model
  - This can be a disadvantage when comparing models
- What is the net effect of adding a new variable?
  - We lose a degree of freedom when a new X variable is added
  - Did the new X variable add enough independent power to offset the loss of one degree of freedom?

Adjusted r^2

- Shows the proportion of variation in Y explained by all X variables adjusted for the number of X variables used

$$r^2 = 1 - \left[(1 - r^2_{Y.12..k}) \left(\frac{n-1}{n-k-1} \right) \right]$$

(where n = sample size, k = number of independent variables)

- Penalizes excessive use of unimportant independent variables
- Smaller than r^2
- Useful in comparing models

Adjusted r^2

Regression Statistics	
Multiple R	0.72213
R Square	0.52148
Adjusted R Square	0.44172
Standard Error	47.46341
Observations	15

$$r^2_{adj} = .44172$$

44.2% of the variation in pie sales is explained by the variation in price and advertising, taking into account the sample size and number of independent variables

ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	2	29460.027	14730.013	6.53861	0.01201
Residual	12	27033.306	2252.776		
Total	14	56493.333			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	306.52619	114.25389	2.68285	0.01993	57.58835	555.46404
Price	-24.97509	10.83213	-2.30565	0.03979	-48.57626	-1.37392
Advertising	74.13096	25.96732	2.85478	0.01449	17.55303	130.70888

F-Test for Overall Significance

- F-Test for Overall Significance of the Model
  - Shows if there is a linear relationship between all of the X variables considered together and Y
- Use F test statistic
  - Hypotheses:

    H_0: $\beta_1 = \beta_2 = \ldots = \beta_k = 0$ (no linear relationship)

    H_1: at least one $\beta_i \neq 0$ (at least one independent variable affects Y)

F-Test for Overall Significance

- Test statistic:

$$F = \frac{MSR}{MSE} = \frac{\frac{SSR}{k}}{\frac{SSE}{n-k-1}}$$

- where F has (numerator) = k and
  (denominator) = (n - k - 1)
  degrees of freedom

F-Test for Overall Significance

Regression Statistics	
Multiple R	0.72213
R Square	0.52148
Adjusted R Square	0.44172
Standard Error	47.46341
Observations	15

$$F = \frac{MSR}{MSE} = \frac{14730.0}{2252.8} = 6.5386$$

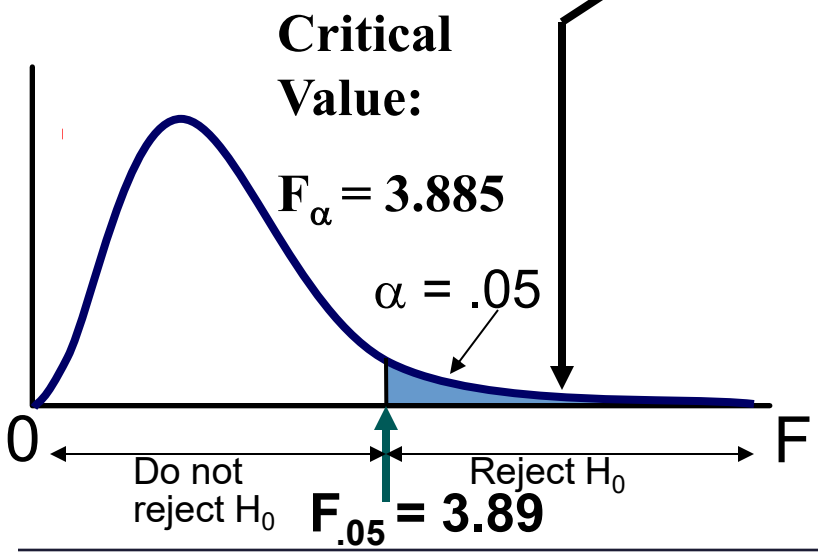
P-value for the F-Test

ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	2	29460.027	14730.013	6.53861	0.01201
Residual	12	27033.306	2252.776		
Total	14	56493.333			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	306.52619	114.25389	2.68285	0.01993	57.58835	555.46404
Price	-24.97509	10.83213	-2.30565	0.03979	-48.57626	-1.37392
Advertising	74.13096	25.96732	2.85478	0.01449	17.55303	130.70888

F-Test for Overall Significance

- H_0: $\beta_1 = \beta_2 = 0$
- H_1: β_1 and β_2 not both zero
- $\alpha = .05$
- $df_1 = 2$ $df_2 = 12$



Test Statistic:

$$F = \frac{MSR}{MSE} = 6.5386$$

Decision:

Since F test statistic is in the rejection region (p-value < .05), reject H_0

Conclusion:

There is evidence that at least one independent variable affects Y

Individual Variables Tests of Hypothesis

- Use t-tests of individual variable slopes
- Shows if there is a linear relationship between the variable X_i and Y
- Hypotheses:

  H_0: $\beta_i = 0$ (no linear relationship)

  H_1: $\beta_i \neq 0$ (linear relationship does exist between X_i and Y)

Individual Variables Tests of Hypothesis

H_0: $\beta_j = 0$ (no linear relationship)

H_1: $\beta_j \neq 0$ (linear relationship does exist between X_i and Y)

- Test Statistic:

$$t = \frac{b_j - 0}{S_{b_j}}$$

(df = n – k – 1)

Individual Variables Tests of Hypothesis

Regression Statistics	
Multiple R	0.72213
R Square	0.52148
Adjusted R Square	0.44172
Standard Error	47.46341
Observations	15

t-value for Price is t = -2.306, with p-value .0398

t-value for Advertising is t = 2.855, with p-value .0145

ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	2	29460.027	14730.013	6.53861	0.01201
Residual	12	27033.306	2252.776		
Total	14	56493.333			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	306.52619	114.25389	2.68285	0.01993	57.58835	555.46404
Price	-24.97509	10.83213	-2.30565	0.03979	-48.57626	-1.37392
Advertising	74.13096	25.96732	2.85478	0.01449	17.55303	130.70888

Individual Variables Tests of Hypothesis

$H_0: \beta_j = 0$

$H_1: \beta_j \neq 0$

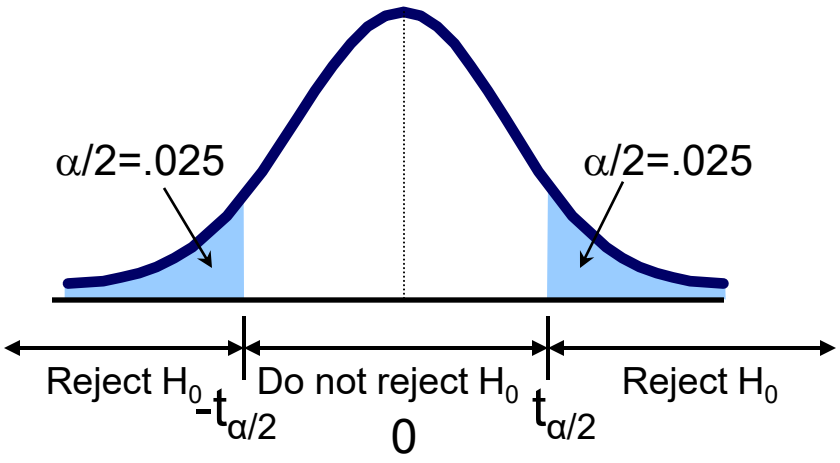
	Coefficients	*Standard Error*	*t Stat*	*P-value*
Price	**-24.97509**	**10.83213**	**-2.30565**	**0.03979**
Advertising	**74.13096**	**25.96732**	**2.85478**	**0.01449**

d.f. = 15 - 2 - 1 = 12

α = .05

$t_{\alpha/2}$ = 2.1788

The test statistic for each variable falls in the rejection region (p-values < .05)



Decision:

Reject H_0 for each variable

Conclusion:

There is evidence that both Price and Advertising affect pie sales at **α** = .05

Confidence Interval Estimate for the Slope

Confidence interval for the population slope β_i

$$b_i \pm t_{n-k-1} S_{b_i}$$

where t has
(n – k – 1) d.f.

	Coefficients	*Standard Error*
Intercept	306.52619	114.25389
Price	-24.97509	10.83213
Advertising	74.13096	25.96732

Here, t has
(15 – 2 – 1) = 12 d.f.

Example: Form a 95% confidence interval for the effect of changes in price (X_1) on pie sales, holding constant the effects of advertising:

-24.975 ± (2.1788)(10.832): So the interval is (-48.576, -1.374)

Confidence Interval Estimate for the Slope

Confidence interval for the population slope β_i

	Coefficients	*Standard Error*	…	*Lower 95%*	*Upper 95%*
Intercept	**306.52619**	**114.25389**	…	**57.58835**	**555.46404**
Price	**-24.97509**	**10.83213**	…	**-48.57626**	**-1.37392**
Advertising	**74.13096**	**25.96732**	…	**17.55303**	**130.70888**

Example: Excel output also reports these interval endpoints:

Weekly sales are estimated to be reduced by between 1.37 to 48.58 pies for each increase of $1 in the selling price, holding constant the effects of advertising.

Using Dummy Variables

- A dummy variable is a categorical independent variable with two levels:
  - yes or no, on or off, male or female
  - coded as 0 or 1
- Assumes equal slopes for other variables
- If more than two levels, the number of dummy variables needed is (number of levels - 1)

Dummy Variable Example

$$\hat{Y} = b_0 + b_1 X_1 + b_2 X_2$$

Let:

Y = pie sales

X_1 = price

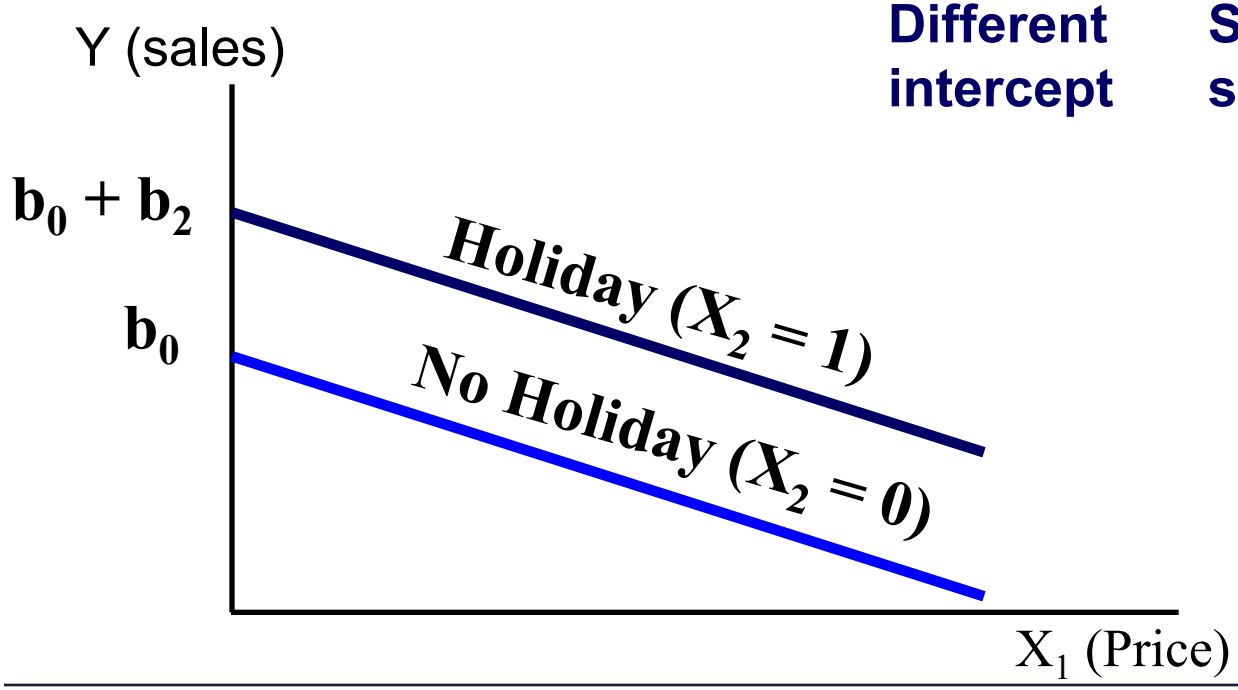
X_2 = holiday (X_2 = 1 if a holiday occurred during the week)
(X_2 = 0 if there was no holiday that week)

Dummy Variable Example

$$\hat{Y} = b_0 + b_1X_1 + b_2(1) = (b_0 + b_2) + b_1X_1 \quad \text{Holiday}$$

$$\hat{Y} = b_0 + b_1X_1 + b_2(0) = b_0 + b_1X_1 \quad \text{No Holiday}$$

Different intercept **Same slope**



If $H_0: \beta_2 = 0$ is rejected, then "Holiday" has a significant effect on pie sales

Dummy Variable Example

$$\text{Sales} = 300 - 30(\text{Price}) + 15(\text{Holiday})$$

Sales: number of pies sold per week

Price: pie price in $

$$\text{Holiday: } \begin{cases} 1 & \text{If a holiday occurred during the week} \\ 0 & \text{If no holiday occurred} \end{cases}$$

$b_2 = 15$: on average, sales were 15 pies greater in weeks with a holiday than in weeks without a holiday, given the same price